

Previously we looked at applications of Geometric Algebra to triangles. We saw in an exercise that spreads can only give a necessary but not sufficient condition for a quadrilateral to be cyclic. In order to use geometric algebra to come up with a measure of the separation between lines which is both 1) definable over any field, and 2) capable of giving a necessary and sufficient condition for a quadrilateral to be cyclic, we need to master arithmetic in our algebra. So far we have only developed theorems in piecemeal, as we need them. In this paper I hope to completely explain aspects of arithmetic in a Geometric Algebra. Many fine documents for this purpose exist, but for my standards none of them are completely rigorous, even the foundational texts on the subject. Here is my attempt at rectifying that. The next paper will return to geometric applications.

For the benefit of the reader, let's recall our axioms and some preliminary results.

A1: We can add together different types of objects We have that $GA(\mathbb{V}, Q(\mathbf{x}))$, written in shorthand as GA , is a ring with unity containing subsets $\mathbb{F} = GA_0$ and $\mathbb{V} = GA_1$, where $\text{char}(\mathbb{F}) \neq 2$.

A2: Addition and multiplication are compatible with scalar arithmetic The ring identities of $GA(\mathbb{V}, Q(\mathbf{x}))$ are the identities of $\mathbb{F}$; $0_{\mathbb{F}}$ and $1_{\mathbb{F}}$. We also have that $\mathbb{F}$ is contained in the center of GA ; scalars commute with everything.

A3: The algebra is generated by vectors Every element in our algebra can be expressed as a linear combination of a product of vectors $GA = \langle \mathbb{V} \rangle$, with the convention that an empty product is the multiplicative identity.

A4: Every vector squares to a scalar For any $\mathbf{x} \in \mathbb{V}$, we have that $\mathbf{x}^2 = Q(\mathbf{x})$, where $Q(\mathbf{x}) = \mathbf{x}^T \Phi \mathbf{x}$ for $\Phi^T = \Phi$ and $\det \Phi \neq 0$.

If $Q(\mathbf{x}) = 0$ our quadratic form sends everything to zero then these axioms are identical to that of the wedge product. With that in mind our first major definition was

$$\mathbf{x}_1 \wedge \dots \wedge \mathbf{x}_k = \frac{1}{k!} \sum_{\sigma \in \mathfrak{S}_k} \text{sign}(\sigma) \mathbf{x}_{\sigma(1)} \dots \mathbf{x}_{\sigma(k)}$$

for the wedge product, which when combined with the fundamental identity

$$\mathbf{a}\mathbf{b} = \frac{1}{2}(\mathbf{a}\mathbf{b} + \mathbf{b}\mathbf{a}) + \frac{1}{2}(\mathbf{a}\mathbf{b} - \mathbf{b}\mathbf{a})$$

allowed us to claim that $\mathbf{a}\mathbf{b} = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b}$. There was a slight hiccup in that when the dimension of our vector space is bigger than the characteristic of our field, we could get 0/0 from our definition of wedge, and that the wedge product is usually seen as a mapping from $\bigwedge^{m_1} \mathbb{F}^n \times \bigwedge^{m_2} \mathbb{F}^n$ to $\bigwedge^{m_1+m_2} \mathbb{F}^n$, not a mapping from $(\bigwedge^1 \mathbb{F}^n)^k$ to $\bigwedge^k \mathbb{F}^n$. To rectify these problems, we let $\mathbf{B} = \mathbf{b}_1 \wedge \dots \wedge \mathbf{b}_k$ and introduced the following formulas

$$\mathbf{a} \wedge \mathbf{B} = \frac{1}{2}(\mathbf{a}\mathbf{B} + (-1)^k \mathbf{B}\mathbf{a}), \quad \mathbf{B} \wedge \mathbf{a} = \frac{1}{2}(\mathbf{B}\mathbf{a} + (-1)^k \mathbf{a}\mathbf{B}).$$

Although we were able to show that this would return us to the associative product we knew and loved, we never actually justified that the formula is consistent with our first definition. This is typical of many expositions about geometric algebra, but anyone can see that it is inadequate to leave things there. Fortunately we can fix this and all other potential problems that can arise in our narrative with a series of definitions and lemmas.

We say that a vector $\mathbf{a} \perp \mathbf{B}$ is perpendicular to blade if $\mathbf{a}$ is perpendicular $\mathbf{a} \cdot \mathbf{b}_i = 0$ to every vector in a factorization of $\mathbf{B} = \mathbf{b}_1 \wedge \dots \wedge \mathbf{b}_k$.

Exercise 1 Show that perpendicularity of a vector to a blade is well defined, it does not depend on the specific factorization of the blade chosen.

We say that a blade is **null** if the restriction of Φ to the subspace that our blade represents is the zero matrix. An example of a null blade is $(\mathbf{a} + \mathbf{c}) \wedge (\mathbf{b} + \mathbf{d})$ where

$$\mathbf{a}^2 = \mathbf{b}^2 = \mathbf{c}^2 = \mathbf{d}^2 = \mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{d} = \mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{d} = 0,$$

$$\mathbf{a} \cdot \mathbf{c} = \mathbf{b} \cdot \mathbf{d} = 1.$$

In the span of $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ we have that $\det \Phi = 1$ but when restricted to the span of $\mathbf{a} + \mathbf{c}$ and $\mathbf{b} + \mathbf{d}$ we have that $\Phi = \mathbf{0}$ our phi matrix is the zero matrix. As we remarked in a previously, the axioms of the geometric product degenerate into the axioms of the wedge product in this subspace, so that $(\mathbf{a} + \mathbf{c})(\mathbf{b} + \mathbf{d}) = (\mathbf{a} + \mathbf{c}) \wedge (\mathbf{b} + \mathbf{d})$. Using this property we can conclude that if $\mathbf{a}$ and $\mathbf{B}$ are a null vector and null blade whose wedge is also null, then we have almost trivially

$$\mathbf{a} \wedge \mathbf{B} = \frac{1}{2}(\mathbf{a}\mathbf{B} + (-1)^k \mathbf{B}\mathbf{a}) = (-1)^n \mathbf{B} \wedge \mathbf{a} = \frac{1}{2}(\mathbf{B}\mathbf{a} + (-1)^k \mathbf{a}\mathbf{B}).$$

It remains to show that this is still true in all other cases.

Lemma 1 Given a vector $\mathbf{a}$ and a non-null blade $\mathbf{B}$ we can always find two unique vectors $\mathbf{a}_{\parallel \mathbf{B}}$ and $\mathbf{a}_{\perp \mathbf{B}}$ such that $\mathbf{a} = \mathbf{a}_{\parallel \mathbf{B}} + \mathbf{a}_{\perp \mathbf{B}}$, with $\mathbf{a}_{\parallel \mathbf{B}} \wedge \mathbf{B} = 0$ and $\mathbf{a}_{\perp \mathbf{B}} \perp \mathbf{B}$.

Proof We can write $\mathbf{a}_{\parallel \mathbf{B}}$ in the form $\mathbf{a}_{\parallel \mathbf{B}} = c_1 \mathbf{b}_1 + \dots + c_k \mathbf{b}_k$ and therefore

$$\mathbf{a}_{\perp \mathbf{B}} = \mathbf{a} - c_1 \mathbf{b}_1 - \dots - c_k \mathbf{b}_k.$$

Using the fact that $\mathbf{a}_{\perp \mathbf{B}}$ is perpendicular to every every $\mathbf{b}_i$ we can write down the matrix system of equations

$$\begin{bmatrix} \mathbf{b}_1 \cdot \mathbf{b}_1 & \dots & \mathbf{b}_1 \cdot \mathbf{b}_k \\ \vdots & \ddots & \vdots \\ \mathbf{b}_k \cdot \mathbf{b}_1 & \dots & \mathbf{b}_k \cdot \mathbf{b}_k \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix} = \begin{bmatrix} \mathbf{a} \cdot \mathbf{b}_1 \\ \vdots \\ \mathbf{a} \cdot \mathbf{b}_k \end{bmatrix}$$

which is guaranteed to be solvable by the non-null assumption on $\mathbf{B}$.

If $\mathbf{B}$ was null then this process can fail in two ways. When $\mathbf{a}$ is already perpendicular to $\mathbf{B}$ then any linear combination of $\mathbf{a}$ and the $\mathbf{b}_i$ s will also be perpendicular to $\mathbf{B}$. Otherwise no linear combination of $\mathbf{a}$ and the $\mathbf{b}_i$ s will be perpendicular to $\mathbf{B}$.

Lemma 2 If Φ restricted to a subspace is invertible, then we can find a basis for that subspace with contains no null vectors, has all basis vectors pairwise perpendicular, and has coefficients in our ground field.

Proof The proof is left to the reader, there are several possible, but the simplest is probably with induction. A constructive proof using completing the square can also be given.

Exercise 2 Find a basis of perpendicular non null vectors for the space spanned by $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ subject to the constraints

$$\mathbf{a}^2 = \mathbf{b}^2 = \mathbf{c}^2 = \mathbf{d}^2 = \mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{d} = \mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{d} = 0,$$

$$\mathbf{a} \cdot \mathbf{c} = \mathbf{b} \cdot \mathbf{d} = 1.$$

Readers might get a little concerned at this point, because the whole idea of geometric algebra is that we do not need to reference a basis to make our computations. While this is true in the big picture, it is often helpful or convenient for us to be able to refactor a blade with terms that have nicer properties. To make our life easier, we define the **grade** of a blade to be the number of terms it factors into. Recall that the sum of blades is not necessarily a blade. In this case we would say that any expression made by summing objects of the same grade k also has that grade k . When we will want to assert a linear combinations of blades is a blade, we will need some sort of algorithm.

Lemma 3 If $\mathbf{A}$ is a non-null blade with grade m_1 and $\mathbf{B}$ is a blade with grade m_2 with the grade of $\mathbf{A} \cap \mathbf{B}$ being m_3 then we can factor $\mathbf{A}$ into m_3 pairwise perpendicular vectors in $\mathbf{A} \cap \mathbf{B}$ and $m_1 - m_3$ vectors not in $\mathbf{A} \cap \mathbf{B}$ which are pairwise perpendicular to each other and our first m_3 vectors.

Proof Use induction and the previous two lemmas.

We will find it useful for the next few results to take a blade and refactoring it into pairwise perpendicular vectors, because it will dramatically simplify our computations.

Lemma 4 If the factors of our non-null blade $\mathbf{b}_i \perp \mathbf{b}_j$ are perpendicular for $i \neq j$ and $i, j = 1, 2, \dots, k$ then we have

$$\mathbf{b}_1 \wedge \dots \wedge \mathbf{b}_k = \mathbf{b}_1 \dots \mathbf{b}_k$$

that we can wedge our factors together by simply multiplying them.

Proof By definition we have that

$$\mathbf{b}_1 \wedge \dots \wedge \mathbf{b}_k = \frac{1}{k!} \sum_{\sigma \in \mathfrak{S}_k} \text{sign}(\sigma) \mathbf{b}_{\sigma(1)} \dots \mathbf{b}_{\sigma(k)}.$$

By rearranging each of the terms in the sum to the identity order we can cancel out the sign of our permutations and recover $k!$ copies of $\mathbf{b}_1 \dots \mathbf{b}_k$, which will be normalized by the $1/(k!)$ in our definition.

There is an important corollary to this lemma which is left as an exercise to the reader.

Exercise 3 Show that if $\mathbf{a}$ is perpendicular to $\mathbf{B}$ then $\mathbf{a} \wedge \mathbf{B} = \mathbf{aB}$.

This is the primary reason that we will refactor our blades, and it allows us to understand the near commutativity of the geometric product in some important cases.

Lemma 5 Given vector $\mathbf{a}$ and non-null blade $\mathbf{B}$ with grade k , define $\mathbf{a}'_{\mathbf{B}} = \mathbf{a}_{\parallel \mathbf{B}} - \mathbf{a}_{\perp \mathbf{B}}$. Then we have

$$\mathbf{a}_{\parallel \mathbf{B}} \mathbf{B} = (-1)^{k-1} \mathbf{B} \mathbf{a}_{\parallel \mathbf{B}}, \quad \mathbf{a}_{\perp \mathbf{B}} \mathbf{B} = (-1)^k \mathbf{B} \mathbf{a}_{\perp \mathbf{B}}, \quad \mathbf{aB} = (-1)^k \mathbf{B} \mathbf{a}'_{\mathbf{B}}.$$

Proof Since $\mathbf{a}_{\parallel \mathbf{B}}$ is in the span of the factors of $\mathbf{B}$ we can refactor $\mathbf{B}$ to be in the form $\mathbf{b}'_1 \wedge \mathbf{b}'_2 \wedge \dots \wedge \mathbf{b}'_k$ for $\mathbf{b}'_1 = \mathbf{a}_{\parallel \mathbf{B}}$ and some other vectors which are pairwise perpendicular to each other and $\mathbf{a}_{\parallel \mathbf{B}}$. We want to drag $\mathbf{a}_{\parallel \mathbf{B}}$ across the factors of $\mathbf{B}$

$$\mathbf{a}_{\parallel \mathbf{B}} \mathbf{B} = \frac{\mathbf{a}_{\parallel \mathbf{B}}}{n!} \sum_{\sigma \in \mathfrak{S}_n} \text{sign}(\sigma) \mathbf{b}'_{\sigma(1)} \dots \mathbf{b}'_{\sigma(k)}.$$

The fundamental identity tells us vectors which are parallel commute, and vectors which are perpendicular anticommute. For us that means when we drag $\mathbf{a}_{\parallel \mathbf{B}}$ across the factors of $\mathbf{B}$ we will pick up $k - 1$ minus signs, so we get

$$\mathbf{a}_{\parallel \mathbf{B}} \mathbf{B} = \left[\frac{(-1)^{k-1}}{n!} \sum_{\sigma \in \mathfrak{S}_n} \text{sign}(\sigma) \mathbf{b}'_{\sigma(1)} \dots \mathbf{b}'_{\sigma(k)} \right] \mathbf{a}_{\parallel \mathbf{B}}$$

as required. Similar logic can prove the second equation, but is even easier to use exercise 3. The third equation follows from adding the first two.

We are within striking distance of our goal, but we need one more piece

Lemma 6 Given a vector $\mathbf{a}$ and a non null blade $\mathbf{B}$ of grade k we have that

$$\mathbf{aB} + (-1)^k \mathbf{Ba} = \mathbf{a}_{\perp \mathbf{B}} \mathbf{B} + (-1)^k \mathbf{Ba}_{\perp \mathbf{B}}, \quad \mathbf{aB} - (-1)^k \mathbf{Ba} = \mathbf{a}_{\parallel \mathbf{B}} \mathbf{B} - (-1)^k \mathbf{Ba}_{\parallel \mathbf{B}}.$$

Proof Now that we know how the non commutativity of vectors with blades works we can write

$$\begin{aligned}\mathbf{aB} &= (-1)^{k-1}\mathbf{Ba}_{\parallel\mathbf{B}} + (-1)^k\mathbf{Ba}_{\perp\mathbf{B}}, \\ \mathbf{Ba} &= (-1)^{k-1}\mathbf{a}_{\parallel\mathbf{B}}\mathbf{B} + (-1)^k\mathbf{a}_{\perp\mathbf{B}}\mathbf{B},\end{aligned}$$

so that our expressions become

$$\begin{aligned}\mathbf{aB} + (-1)^n\mathbf{Ba} &= (-1)^{k-1}\mathbf{Ba}_{\parallel\mathbf{B}} + (-1)^k\mathbf{Ba}_{\perp\mathbf{B}} - \mathbf{a}_{\parallel\mathbf{B}}\mathbf{B} + \mathbf{a}_{\perp\mathbf{B}}\mathbf{B}, \\ \mathbf{aB} - (-1)^k\mathbf{Ba} &= (-1)^{k-1}\mathbf{Ba}_{\parallel\mathbf{B}} + (-1)^k\mathbf{Ba}_{\perp\mathbf{B}} + \mathbf{a}_{\parallel\mathbf{B}}\mathbf{B} - \mathbf{a}_{\perp\mathbf{B}}\mathbf{B}.\end{aligned}$$

We can move cancel out the parts of $\mathbf{a}$ parallel to $\mathbf{B}$ in the first equation, and cancel out the parts that are perpendicular in the second, achieving the results.

Now we can prove our formula for wedge in the case that $\mathbf{B}$ is not a null blade using the lemmas we have collected so far.

Theorem 1 If $\mathbf{a}$ is a vector and $\mathbf{B}$ is a non null blade of grade k , then we can write

$$\mathbf{a} \wedge \mathbf{B} = \frac{1}{2}(\mathbf{aB} + (-1)^k\mathbf{Ba}).$$

Proof Using all of our previous lemmas we can say

$$\mathbf{a} \wedge \mathbf{B} = \mathbf{a}_{\perp\mathbf{B}} \wedge \mathbf{B} = \mathbf{a}_{\perp\mathbf{B}}\mathbf{B} = \frac{1}{2}(\mathbf{a}_{\perp\mathbf{B}}\mathbf{B} + (-1)^k\mathbf{Ba}_{\perp\mathbf{B}}) = \frac{1}{2}(\mathbf{aB} + (-1)^k\mathbf{Ba})$$

as required.

With this formula in hand we have opened the door to understanding the structure of the geometric product.

Exercise 4 There is technically one more case we need to consider before we can move on, suppose we have a non-null vector which is not perpendicular to a null vector. Show that the formula still holds in this case. Show that the case of a null vector which is not perpendicular to a null blade can be reduced to the previous situation. Conclude that if $\mathbf{B} = \mathbf{b}_1 \wedge \dots \wedge \mathbf{b}_k$ then

$$\mathbf{aB} + (-1)^k\mathbf{Ba} = (\mathbf{a} + x_1\mathbf{b}_1 + \dots + x_k\mathbf{b}_k)\mathbf{B} + (-1)^k\mathbf{B}(\mathbf{a} + x_1\mathbf{b}_1 + \dots + x_k\mathbf{b}_k)$$

for any scalars $x_1, \dots, x_k \in \mathbb{F}$.

Using the identity $(\mathbf{a} \cdot \mathbf{c})\mathbf{B} = \mathbf{B}(\mathbf{a} \cdot \mathbf{c})$ we can prove $(\mathbf{a} \wedge \mathbf{B}) \wedge \mathbf{c} = \mathbf{a} \wedge (\mathbf{B} \wedge \mathbf{c})$ and thus recursively establish associativity, as was done in an exercise of a previous essay. We have briefly commented on the fact that it is possible for formal linear combinations of blades to not be blades themselves.

The prototypical example would be $\mathbf{b}_1 \wedge \mathbf{b}_2 + \mathbf{b}_3 \wedge \mathbf{b}_4$ which cannot be factored into a wedge of two vectors not matter which quadratic form we try to impose on our space, or field we chose to work over. But how do we know that for sure?

We would like an algorithm to determine if a sum of k blades is itself a k blade. To this end we define $W_{\mathbf{B}}$ the **form envelope** of a sum of k blades $\mathbf{B} = \mathbf{B}_1 + \mathbf{B}_2 + \dots + \mathbf{B}_n$ to be the vector space of solutions to the equation $\mathbf{x} \wedge \mathbf{B} = 0$. We can use the dimension of this space to determine if an expression is or is not a blade.

Lemma 7 A sum of k blades $\mathbf{B} = \mathbf{B}_1 + \mathbf{B}_2 + \dots + \mathbf{B}_n$ is itself also a blade if and only if $\dim(W_{\mathbf{B}}) = k$.

Proof ($\Leftarrow$) Using the fact that $\mathbf{a} \wedge \mathbf{B} = \mathbf{0}$ if and only if $\mathbf{B} = \mathbf{a} \wedge \mathbf{C}$ for some $k - 1$ blade $\mathbf{C}$, a basis for $W_{\mathbf{B}}$ can be wedged together into a k -blade which is multiple of our expression. ($\Rightarrow$) Now suppose that $\mathbf{B} = \mathbf{b}_1 \wedge \dots \wedge \mathbf{b}_k$ is a k blade. Then $\mathbf{a} \wedge \mathbf{B} = 0$ means that $\mathbf{a} = x_1 \mathbf{b}_1 + \dots + x_k \mathbf{b}_k$ which has k free parameters as required.

Exercise 5 Let $\mathbf{B} = \mathbf{b}_1 \wedge \mathbf{b}_2 + \mathbf{b}_3 \wedge \mathbf{b}_4$ and show $\dim(W_{\mathbf{B}}) = 0$. For a set of k vectors $\mathbf{b}_1, \dots, \mathbf{b}_k$ take $\mathbf{b}_1 \wedge \dots \wedge \check{\mathbf{b}}_i \wedge \dots \wedge \mathbf{b}_k$ to mean all the vectors except $\mathbf{b}_i$ wedged together. Show that

$$\sum_{i=1}^k c_i \mathbf{b}_1 \wedge \dots \wedge \check{\mathbf{b}}_i \wedge \dots \wedge \mathbf{b}_k$$

is always a $k - 1$ blade for any numbers $c_1, \dots, c_k$.

At this point we have established that we can faithfully recover our familiar wedge product from the geometric product. We can extend our fundamental identity to read

$$\mathbf{a}\mathbf{B} = \frac{1}{2}(\mathbf{a}\mathbf{B} - (-1)^k \mathbf{B}\mathbf{a}) + \mathbf{a} \wedge \mathbf{B}$$

$$\mathbf{B}\mathbf{a} = \frac{1}{2}(\mathbf{B}\mathbf{a} - (-1)^k \mathbf{a}\mathbf{B}) + \mathbf{B} \wedge \mathbf{a}$$

but this begs the question: what is the geometric meaning of $\frac{1}{2}(\mathbf{a}\mathbf{B} - (-1)^k \mathbf{B}\mathbf{a})$ as well as $\frac{1}{2}(\mathbf{B}\mathbf{a} - (-1)^k \mathbf{a}\mathbf{B})$? To answer this question we define two new symbols called the **right and left contractions**

$$\mathbf{a} \rfloor \mathbf{B} = \frac{1}{2}(\mathbf{a}\mathbf{B} - (-1)^k \mathbf{B}\mathbf{a}), \quad \mathbf{B} \llbracket \mathbf{a} = \frac{1}{2}(\mathbf{B}\mathbf{a} - (-1)^k \mathbf{a}\mathbf{B})$$

which are not always commutative like the dot product on vectors, but instead are related by $\mathbf{a} \rfloor \mathbf{B} = (-1)^{k+1} \mathbf{B} \llbracket \mathbf{a}$. If we let $\mathbf{B}$ be a vector itself then we recover our original formula for the dot product, but we will need more lemmas if we want to proceed. We have gotten very far just by appealing to geometry, but a little knowledge of group theory will give us a lemma which suggests alternate routes for all the results we have generated so far.

Lemma 8 For a vector $\mathbf{a}$ and a blade $\mathbf{B} = \mathbf{b}_1 \wedge \dots \wedge \mathbf{b}_k$ we can

$$\mathbf{aB} = \sum_{i=1}^k (-1)^{i+1} (\mathbf{a} \mathbf{b}_i) \mathbf{b}_1 \wedge \dots \wedge \check{\mathbf{b}}_i \wedge \dots \wedge \mathbf{b}_k, \quad \mathbf{Ba} = \sum_{i=1}^k (-1)^{n-i} \mathbf{b}_1 \wedge \dots \wedge \check{\mathbf{b}}_i \wedge \dots \wedge \mathbf{b}_k (\mathbf{b}_i \mathbf{a})$$

peel off the factors of $\mathbf{B}$ one at a time to multiply them by $\mathbf{a}$ separately.

Proof We will prove the first equality, the proof of the other follows the same logic. By definition we have

$$\mathbf{aB} = \frac{\mathbf{a}}{k!} \sum_{\sigma \in \mathfrak{S}_k} \text{sign}(\sigma) \mathbf{b}_{\sigma(1)} \dots \mathbf{b}_{\sigma(k)}$$

To move forward we will look at every permutation that brings $\mathbf{b}_i$ to the leftmost position adjacent to $\mathbf{a}$. For convenience we define $\mathfrak{S}_{k/i}$ a set of functions τ which act on $\tau(\mathbf{b}_j) = \mathbf{b}_{\tau(j)}$ vectors by permuting their index if $i \neq j$, and $\tau(\mathbf{b}_i) = \mathbf{b}_{\tau(i)} = 1$ send it to one otherwise. Every $\tau \in \mathfrak{S}_{k/i}$ can be obtained from k distinct $\sigma \in \mathfrak{S}_k$ which move $\mathbf{b}_i$ to the left (or right). If our functions $\tau \in \mathfrak{S}_{k/i}$ act on a set of $k-1$ vectors indexed from $1, \dots, \check{i}, \dots, k$ then it becomes a group again, where the sign of each function is $(-1)^{i+1}$ times what it would be as a permutation $\sigma \in \mathfrak{S}_k$ which fixed $\mathbf{b}_i$. Using these considerations we can conclude that

$$\mathbf{b}_1 \wedge \dots \wedge \check{\mathbf{b}}_i \wedge \dots \wedge \mathbf{b}_k = \frac{1}{(k-1)!} \sum_{\tau \in \mathfrak{S}_{k/i}} \text{sign}(\tau) \mathbf{b}_{\tau(1)} \dots \mathbf{b}_{\tau(i)} \dots \mathbf{b}_{\tau(k)}$$

and so

$$\mathbf{aB} = \frac{\mathbf{a}}{k!} \sum_{i=1}^k k (-1)^{i+1} \mathbf{b}_i \sum_{\tau \in \mathfrak{S}_{k/i}} \text{sign}(\tau) \mathbf{b}_{\tau(1)} \dots \mathbf{b}_{\tau(i)} \dots \mathbf{b}_{\tau(k)}$$

which gives us the result after simplifying.

Exercise 6 Use lemma 8 to prove the following three expansions of $\wedge, \rfloor, \lrcorner$

$$\mathbf{a} \wedge \mathbf{B} = \frac{1}{2} \sum_{i=1}^k [(-1)^{i+1} (\mathbf{a} \wedge \mathbf{b}_i) (\mathbf{b}_1 \wedge \dots \wedge \check{\mathbf{b}}_i \wedge \dots \wedge \mathbf{b}_k) + (-1)^{k-i} (\mathbf{b}_1 \wedge \dots \wedge \check{\mathbf{b}}_i \wedge \dots \wedge \mathbf{b}_k) (\mathbf{b}_i \wedge \mathbf{a})],$$

$$\mathbf{a} \rfloor \mathbf{B} = \sum_{i=1}^k (-1)^{i+1} (\mathbf{a} \cdot \mathbf{b}_i) (\mathbf{b}_1 \wedge \dots \wedge \check{\mathbf{b}}_i \wedge \dots \wedge \mathbf{b}_k),$$

$$\mathbf{B} \lrcorner \mathbf{a} = \sum_{i=1}^k (-1)^{k-i} (\mathbf{a} \cdot \mathbf{b}_i) (\mathbf{b}_1 \wedge \dots \wedge \check{\mathbf{b}}_i \wedge \dots \wedge \mathbf{b}_k),$$

and conclude that contracting a vector onto a k blade produces a $k-1$ blade, contained in the k blade.

Lemma 6 says that our contractions only depend on the part of $\mathbf{a}$ which is contained in $\mathbf{B}$, so that if $\mathbf{a}$ is perpendicular to $\mathbf{B}$ then contracting it to either side of $\mathbf{B}$ produces 0. Therefore our geometric interpretations of $\mathbf{a} \rfloor \mathbf{B}$ and $\mathbf{B} \rfloor \mathbf{a}$ are that they are representatives of $\mathbf{B} \perp_{\mathbf{a}}$ the part of $\mathbf{B}$ most unlike $\mathbf{a}$ from the perspective of our quadratic form. A special case of our formula informs us that vectors are perpendicular to all scalars, and therefore scalars are perpendicular to all non-scalar blades.

It makes sense to consider the contractions as fully fledged products, because they distribute over both arguments. Unlike the wedge product, our definition of contraction is not associative, so that $(\mathbf{a} \rfloor \mathbf{b}) \rfloor \mathbf{C} \neq \mathbf{a} \rfloor (\mathbf{b} \rfloor \mathbf{C})$ and $(\mathbf{C} \rfloor \mathbf{b}) \rfloor \mathbf{a} \neq \mathbf{C} \rfloor (\mathbf{b} \rfloor \mathbf{a})$ because we have not even said what it means to contract a scalar onto a blade (once we do we can see more conclusively that contractions are not associative). Despite this, we could show $\mathbf{a} \rfloor (\mathbf{B} \rfloor \mathbf{c}) = (\mathbf{a} \rfloor \mathbf{B}) \rfloor \mathbf{c}$ using the same technique we used to prove associativity for wedge.

Exercise 7 Now that we have three derived products floating around, we want to understand how they interact with each other. We will define our order of operations to favor $\wedge$ first, then $\rfloor$, $\rfloor$, and finally our geometric product last. For a vector $\mathbf{a}$, a k_1 blade $\mathbf{B}$, and a k_2 blade $\mathbf{C}$, show that

$$\begin{aligned}\mathbf{a} \rfloor (\mathbf{B} \wedge \mathbf{C}) &= (\mathbf{a} \rfloor \mathbf{B}) \wedge \mathbf{C} + (-1)^{k_1} \mathbf{B} \wedge (\mathbf{a} \rfloor \mathbf{C}) \\ (\mathbf{B} \wedge \mathbf{C}) \rfloor \mathbf{a} &= (-1)^{k_2} (\mathbf{B} \rfloor \mathbf{a}) \wedge \mathbf{C} + \mathbf{B} \wedge (\mathbf{C} \rfloor \mathbf{a})\end{aligned}$$

we can graded distribute our contractions over wedge.

So far we have accomplished quite a lot, we can now state the **extended fundamental identities**

$$\mathbf{a} \mathbf{B} = \mathbf{a} \rfloor \mathbf{B} + \mathbf{a} \wedge \mathbf{B}, \quad \mathbf{B} \mathbf{a} = \mathbf{B} \rfloor \mathbf{a} + \mathbf{B} \wedge \mathbf{a}$$

which tell us that the geometric product of a vector with a blade is the sum of a $k - 1$ blade representing the "difference" of the spaces spanned by the vector and the blade plus a $k + 1$ blade representing the "sum" of the spaces spanned by the vector and the blade. With this in mind we know that repeated applications of the geometric product cannot be used to construct anything other than blades, and our geometric algebra is a direct sum of the subspaces $GA_0(\mathbb{F}^n, Q(\mathbf{x})) = \mathbb{F}$, $GA_1(\mathbb{F}^n, Q(\mathbf{x})) = \mathbb{F}^n$, $GA_2(\mathbb{F}^n, Q(\mathbf{x})) = \bigwedge^2 \mathbb{F}^n$, ... , $GA_n(\mathbb{F}^n, Q(\mathbf{x})) = \bigwedge^n \mathbb{F}^n$.

It is not quite correct to call this algebra graded because the product of two elements take values in multiple distinct subspaces, but it is fair to say that this algebra is multigraded. To help us study how this multigrading works, we define the **grade projection** operators $\langle \rangle_k$ which project a multivector $\mathbf{M} = m_0 + \mathbf{m}_1 + \mathbf{m}_2 \wedge \mathbf{m}_3 + \dots$ onto its k blade part so that $\langle \mathbf{B} \rangle_0 = m_0$ and $\langle \mathbf{M} \rangle_k = \mathbf{m}_{\frac{1}{2}(k+1)k} \wedge \dots \wedge \mathbf{m}_{\frac{1}{2}(k+2)(k+1)}$. Since there are no blades of negative grade we take grade projections with negative index to send everything to 0. As suggested by their names, our grade projections are linear and idempotent.

Using induction we can see that for a blade $\mathbf{B}$ of grade m_1 and a blade $\mathbf{C}$ of grade m_2 , the product $\mathbf{BC}$ can take nonzero values in the subspaces GA_m , GA_{m+2} , GA_{m+4} , all the way up to $GA_{m_1+m_2}$, where $m = \max(m_1, m_2) - \min(m_1, m_2)$. The subspaces which take nonzero values depend on of how relatively parallel and how relatively perpendicular they are.

Exercise 8 The first nontrivial example of this multi grading is found in the expansion of $(\mathbf{a} \wedge \mathbf{b})(\mathbf{c} \wedge \mathbf{d})$. Show that we have

$$\begin{aligned} (\mathbf{a} \wedge \mathbf{b})(\mathbf{c} \wedge \mathbf{d}) = & (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c}) - (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) \\ & + (\mathbf{b} \cdot \mathbf{c})\mathbf{a} \wedge \mathbf{d} - (\mathbf{b} \cdot \mathbf{d})\mathbf{a} \wedge \mathbf{c} \\ & + (\mathbf{a} \cdot \mathbf{d})\mathbf{b} \wedge \mathbf{c} - (\mathbf{a} \cdot \mathbf{c})\mathbf{b} \wedge \mathbf{d} \\ & + \mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c} \wedge \mathbf{d} \end{aligned}$$

that the product of two general bivectors is a scalar plus bivector plus quadvector. The rest of this exercise is not needed for the rest of the exposition, but is well within the theme. For two bivectors $\mathbf{A}$ and $\mathbf{B}$ define $\mathbf{A} \times \mathbf{B} = \frac{1}{2}(\mathbf{AB} - \mathbf{BA})$ the **commutator** of $\mathbf{A}$ and $\mathbf{B}$. Show that the commutator of two bivectors is itself a bivector. Show that the commutator is not associative, but it does satisfy the jacobi identity

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) + \mathbf{B} \times (\mathbf{C} \times \mathbf{A}) + \mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = 0$$

making $(GA_2(\mathbb{F}, Q(\mathbf{x})), \times)$ a lie subalgebra of $GA(\mathbb{F}, Q(\mathbf{x}))$.

In order to extend the meaning of our wedge and contraction products to include blades for all arguments in a way consistent with our previous definitions, we will use the grade projection operators. For a blade $\mathbf{B}$ of grade m_1 and a blade $\mathbf{C}$ of grade m_2 we observe that $\mathbf{A} \wedge \mathbf{B} = \langle \mathbf{AB} \rangle_{m_1+m_2}$. Similarly we define $\mathbf{A} \rfloor \mathbf{B} = \langle \mathbf{AB} \rangle_{m_2-m_1}$ and $\mathbf{A} \llcorner \mathbf{B} = \langle \mathbf{AB} \rangle_{m_1-m_2}$. From this definition it is clear that for a scalar a and a non scalar blade $\mathbf{B}$ we have that $a \rfloor \mathbf{B} = a\mathbf{B}$ and that $a \llcorner \mathbf{B} = 0$. What is less clear is the next theorem.

Theorem 2 For blades $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ of grades m_1 , m_2 , and m_3 , we have that

$$(\mathbf{A} \wedge \mathbf{B}) \rfloor \mathbf{C} = \mathbf{A} \rfloor (\mathbf{B} \rfloor \mathbf{C}), \quad \mathbf{C} \llcorner (\mathbf{B} \wedge \mathbf{A}) = (\mathbf{C} \llcorner \mathbf{B}) \llcorner \mathbf{A}$$

which lets us contract a blade onto another by contracting with its factors one at a time.

Proof We will prove the first equality, the logic for the second is identical. Both sides are zero unless $m_1 + m_2 \leq m_3$ in which case

$$(\mathbf{A} \wedge \mathbf{B}) \rfloor \mathbf{C} = \langle (\mathbf{A} \wedge \mathbf{B}) \mathbf{C} \rangle_{m_3-m_1-m_2} = \langle \langle \mathbf{AB} \rangle_{m_1+m_2} \mathbf{C} \rangle_{m_3-m_1-m_2}$$

Since $\langle \mathbf{AB} \rangle_{m_1+m_2}$ is the only part of $\mathbf{AB}$ that can contribute to the $m_3 - m_2 - m_1$ part of $\mathbf{ABC}$ we have that $\langle \langle \mathbf{AB} \rangle_{m_1+m_2} \mathbf{C} \rangle_{m_3-m_1-m_2} = \langle \mathbf{ABC} \rangle_{m_3-m_1-m_2}$. Now since $\langle \mathbf{BC} \rangle_{m_3-m_2}$ is the only part of $\mathbf{BC}$ that contributes to $\langle \mathbf{ABC} \rangle_{m_3-m_1-m_2}$, then we can write

$$\langle \mathbf{ABC} \rangle_{m_3-m_1-m_2} = \langle \mathbf{A} \langle \mathbf{BC} \rangle_{m_3-m_2} \rangle_{m_3-m_1-m_2} = \mathbf{A} \rfloor \langle \mathbf{BC} \rangle_{m_3-m_2} = \mathbf{A} \rfloor (\mathbf{B} \rfloor \mathbf{C})$$

as required.

With this theorem in hand we have almost completed our discussion of the derived products of the geometric algebra, and can move back into pure geometry. The last thing we need to do is crucial for that goal. We define the **scalar product** of two blades $\mathbf{A}$ and $\mathbf{B}$ as $\mathbf{A} * \mathbf{B} = \langle \mathbf{AB} \rangle_0$ the scalar part of their product. This expression is linear in both arguments, so we are justified in calling it a product, however, it is clearly not associative unless all our blades are scalars, in which case it turns back into field multiplication. We can see that this product is zero unless both blades have the same grade. For our final theorem we will find a formula for the scalar product in terms of the factors of our two blades.

Theorem 3 Given k blades $\mathbf{A} = \mathbf{a}_1 \wedge \dots \wedge \mathbf{a}_k$ and $\mathbf{B} = \mathbf{b}_1 \wedge \dots \wedge \mathbf{b}_k$ we have that

$$\mathbf{A} * \mathbf{B} = \det \begin{bmatrix} \mathbf{a}_1 \cdot \mathbf{b}_k & \mathbf{a}_1 \cdot \mathbf{b}_{k-1} & \dots & \mathbf{a}_1 \cdot \mathbf{b}_1 \\ \mathbf{a}_2 \cdot \mathbf{b}_k & \mathbf{a}_2 \cdot \mathbf{b}_{k-1} & \dots & \mathbf{a}_2 \cdot \mathbf{b}_1 \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_k \cdot \mathbf{b}_k & \mathbf{a}_k \cdot \mathbf{b}_{k-1} & \dots & \mathbf{a}_k \cdot \mathbf{b}_1 \end{bmatrix}$$

their scalar product is the determinant of their **gram matrix**.

Proof The proof is by induction on the grade of the blades. When our blades are scalar this definition is trivially equal to scalar multiplication, so our base case is established. Now for our induction hypothesis assume it is true for $k - 1$ blades and write $\mathbf{A} = \mathbf{A}' \wedge \mathbf{a}_k$ and $\mathbf{B} = \mathbf{b}_1 \wedge \mathbf{B}'$ for $k - 1$ blades $\mathbf{A}'$ and $\mathbf{B}'$. Now note that when the grades of two blades are equal that $\mathbf{A} * \mathbf{B} = \mathbf{A} \rfloor \mathbf{B} = \mathbf{A} \lrcorner \mathbf{B}$, so that

$$\mathbf{A} * \mathbf{B} = (\mathbf{A} \wedge \mathbf{a}_k) * (\mathbf{b}_1 \wedge \mathbf{B}') = (\mathbf{A} \wedge \mathbf{a}_k) \rfloor (\mathbf{b}_1 \wedge \mathbf{B}') = (\mathbf{a}_k \cdot \mathbf{b}_1) \mathbf{A}' \rfloor \mathbf{B}' - \mathbf{A}' \rfloor (\mathbf{b}_1 \wedge (\mathbf{a}_k \rfloor \mathbf{B}')).$$

By turning the right contractions back into scalar products we have

$$\mathbf{A} * \mathbf{B} = (\mathbf{a}_k \cdot \mathbf{b}_1) \mathbf{A}' * \mathbf{B}' - \mathbf{A}' * (\mathbf{b}_1 \wedge (\mathbf{a}_k \rfloor \mathbf{B}'))$$

and by expanding the $\mathbf{a}_k \rfloor \mathbf{B}'$ term into a sum of $k - 1$ blades we have

$$\mathbf{A} * \mathbf{B} = \sum_{i=1}^k (-1)^{i+1} (\mathbf{a}_k \cdot \mathbf{b}_i) \mathbf{A}' * (\mathbf{b}_1 \wedge \dots \wedge \check{\mathbf{b}}_i \wedge \dots \wedge \mathbf{b}_k)$$

which, according to our induction hypothesis, is the Laplace expansion of the determinant for the gram matrix between $\mathbf{A}$ and $\mathbf{B}$ along the last row. This establishes the result.

Exercise 9 Define the **quadrance** of a blade $\mathbf{B}$ to be $Q(\mathbf{B}) = \mathbf{B} * \mathbf{B}^\dagger$ where $\mathbf{B}^\dagger = (-1)^{\frac{1}{2}(k^2+k)} \mathbf{B}$ is the blade with its factors wedged in reverse. Write out the formula for the quadrance of a bivector. How is it related to the cross law from rational trigonometry? What does the Cauchy Binet theorem for determinants tell us about the relationship between the quadrance of a blade and the quadrances of our basis blades?